

Celeste T. Pallone, Ph.D.

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EDUCATION

Columbia University New York, NY
M.A., M.Phil., Ph.D., Earth and Environmental Sciences 2025
Dissertation: *Multi-Proxy Paleoceanographic Reconstructions of the Late Pleistocene Eastern Equatorial Pacific*
Advisor: Dr. Jerry F. McManus
Dissertation Committee: Drs. Gisela Winckler, Braddock Linsley, Richard Seager, and
Kassandra Costa (WHOI)

Barnard College New York, NY
B.A., *magna cum laude*, Environmental Science 2019

RESEARCH EXPERIENCE

Massachusetts Institute of Technology Cambridge, MA
Postdoctoral Associate, Dept. of Earth, Atmospheric & Planetary Sciences 2024-Present

- Preparing deep-sea sediments for He isotope analysis to reconstruct the timescales of early Eocene hyperthermal events
- Analyzed major/minor/trace elements in deep-sea sediments to reconstruct equatorial Atlantic bottom water redox state across early Eocene hyperthermal events
- Advisor: Dr. David McGee

Columbia University New York, NY
Graduate Research Assistant, Lamont-Doherty Earth Observatory (LDEO) 2019-2024

- Reconstructed water column structure and sea-surface variability in the late Pleistocene eastern equatorial Pacific using stable isotope analyses of planktic foraminifera
- Analyzed U-Th-Pa isotopes in deep-sea sediments to reconstruct paleo-productivity and carbon cycling in the late Pleistocene eastern equatorial Pacific
- Advisor: Dr. Jerry F. McManus

Undergraduate Research Assistant, LDEO 2018-2019

- Characterized surface sediment properties in the eastern Long Island Sound
- Advisors: Dr. Timothy C. Kenna and Dr. Frank Nitsche

PUBLICATIONS

**Mentored student co-authors*

Chen, X., et al., (including **Pallone, C.T.**). (2026). Tracking variability in atmospheric, oceanic, and riverine inputs to the Iberian Margin over the last 250 kyr. *Paleoceanography and Paleoclimatology*, 41, e2026PA005431. <https://doi.org/10.1029/2026PA005431>.

Pallone, C.T., Holmes, K.,* Morris, C.A.,* McManus, J.F. (2026). Orbital forcing of the eastern equatorial Pacific thermocline in the late Pleistocene. *Quaternary Science Reviews*, 373, 109734. <https://doi.org/10.1016/j.quascirev.2025.109734>.

Hodell, D.A., et al. (including **Pallone, C.T.**). (2026). Onset of millennial climate variability with the intensification of Northern Hemisphere glaciation. *Science*, 392, 57-62.

<https://doi.org/10.1126/science.ady7970>.

Zhou, Y.Z., McManus, J.F., **Pallone, C.T.**, Kenna, T.C., Weinstein, G.A., Garcia, H. (2025). Abrupt weakening of deep Atlantic circulation at the last glacial inception. *Nature Communications*, 16:7555. <https://doi.org/10.1038/s41467-025-62960-y>.

Jacobel, A.W., **Pallone, C.T.**, Costa, K.M., Anderson, R.F., McManus, J.F. (2025). Orbital Influences on Deep Ocean Oxygen Concentrations and Respired Carbon Storage. *Global Biogeochemical Cycles*, 39, e2025GB008503. <https://doi.org/10.1029/2025GB008503>.

FUTURE 2024 PI-team, et al. (including **Pallone, C.T.**) (2025). The FUTURE of the US marine seafloor & subseafloor sampling capabilities. *AGU Advances*, 6, e2024AV001560. <https://doi.org/10.1029/2024AV001560>.

Pallone, C.T., McManus, J.F., Jacobel, A.W. (2025). Eastern equatorial Pacific paleo-productivity and carbon cycling during the late Pleistocene. *Earth and Planetary Science Letters*, 656, 119255. <https://doi.org/10.1016/j.epsl.2025.119255>.

Co-author of 8 Chapters & Data Reports in: Hodell, D.A., Abrantes, F., Alvarez Zarikian, C.A., **the Expedition 397 Scientists** (2024). Iberian Margin Paleoclimate. *Proceedings of the International Ocean Discovery Program (IODP)*, 397. IODP. <https://doi.org/10.14379/iodp.proc.397.2024>.

Hodell, D.A., Abrantes, F., Alvarez Zarikian, C.A., **the Expedition 397 Scientists** (2023). Expedition 397 Preliminary Report: Iberian Margin Paleoclimate. IODP. <https://doi.org/10.14379/iodp.pr.397.2023>.

In Development

Pallone, C.T., McManus, J.F., Morris, C.A.,* Ejupi, A.* Evaluating Orbital Forcing of ENSO with Individual Foraminifera Stable Isotope Analysis of Eastern Equatorial Pacific *Globigerinoides ruber* from Marine Isotope Stages 5 and 6. In revision at *Paleoceanography and Paleoclimatology*.

Zhou, Y.Z., Lakhani, K.Q., Lynch-Stieglitz, J., Gebbie, G., Findley, B., **Pallone, C.T.**, O'Keefe, A.G.,* Liu, H. Deeper thermocline indicates stronger Pacific Walker circulation during the Last Glacial Maximum. Under Review at *Science*.

HONORS & AWARDS

Grace Potter Rice Prize in the Natural Sciences, Barnard College	2019
Anne Davidson Prize & Departmental Honors, Environmental Science, Barnard College	2019
Phi Beta Kappa, Barnard College	2018

GRANTS & FELLOWSHIPS

Schlanger Fellowship, IODP-U.S. Science Support Program (USSSP) (\$30,000)	2023-2024
• <i>Paleo-Productivity in the Late Pleistocene Eastern Equatorial Pacific</i>	
Post-Expedition Activity Award, IODP-USSSP (\$20,000)	2023-2024
• <i>Uranium Series and Benthic Foraminifera Stable Isotope Analysis of IODP Expedition 397 Sites U1385 and U1587 over Late Pleistocene Glacial Cycles</i>	
• Dr. Jerry F. McManus as Administrative PI	
Graduate Student Travel Grant, Columbia University (\$850)	2021

Fall Meeting Student Travel Grant, American Geophysical Union (AGU) (\$500)	2018
Student Travel Grant, Earth Institute, Columbia University (\$750)	2017

SEA-GOING EXPERIENCE

International Ocean Discovery Program, R/V JOIDES Resolution <i>Sedimentologist, Expedition 397: Iberian Margin Paleoclimate</i>	2022
Long Island Sound Mapping Project, R/V Seawolf & R/V Pritchard <i>Undergraduate Research Assistant, LIS 1802 & 1803</i>	2018
Sea Education Association, SSV Corwith Cramer <i>Student, C-276: Caribbean Reef Expedition</i>	2017

TEACHING EXPERIENCE

Columbia University, Dept. of Earth & Environmental Sciences	New York, NY
<i>Course Assistant</i>	
Climate Systems (EESC 2100)	Spring 2024
- Designed undergraduate-level lab lectures, activities, and assignments in collaboration with faculty and TA team for 50+ students - Facilitated hands-on lab exercises for 15-student section	
<i>Guest Lecturer</i>	
Terrestrial Paleoclimate (EESC 4330)	Fall 2023
Developed and delivered graduate-level lecture on ocean circulation and paleoclimate	
Climate Systems (EESC 2100)	Spring 2022, Spring 2023
Developed and delivered 4 undergraduate-level lectures on ocean circulation, ocean chemistry, and paleoclimate	
<i>Graduate Teaching Assistant</i>	
Climate Systems (EESC 2100)	Fall 2021
- Designed and led lab lectures, activities, and assignments in climate and ocean sciences for 50 undergraduate students, including Earth's energy balance, ocean circulation, and paleoclimate proxy records - Graded lab assignments and held weekly office hours - Coordinated with faculty and TA team on curriculum planning and student assessment	
Field Geology: Barbados (EESC 3010)	Spring 2020
- Coached 20 students in developing and delivering presentations on geology, paleoclimate, hydrology, and coral reef ecology of Barbados - When field component was cancelled due to the pandemic, supported students remotely in creating and presenting field guides for local geology	
<i>Undergraduate Teaching Assistant</i>	
Solid Earth Systems (EESC 2200)	Spring 2019
Environmental Data Analysis (EESC 3017)	Fall 2018

MENTORSHIP

Massachusetts Institute of Technology	Cambridge, MA
<i>Undergraduate Research Opportunities Program</i>	
Reeya Duvvuri	2026
Sofia Weber	2025-2026
Columbia University	New York, NY
<i>LDEO Research Experiences for Undergraduates</i>	
Cire Portegies	2024
Martin Aguilar Solano	2024
Christopher Cumbe	2024
Naomi Bonilla	2024
Sydney Smith	2023
Now: Ph.D. Student at Columbia University	
Erin Kim	2023
Now: Ph.D. Candidate at University of Cambridge	
Amanda Perrone	2023
Cidney McMahon	2023
<i>Co-Advised Undergraduate Senior Theses</i>	
Macy Matthews, Columbia University	2023-2024
Helen Latting, Columbia University	2023-2024
Now: 2025-2026 Fulbright Scholar at University of Bergen	
Isabel Plower, Columbia University	2023-2024
Alexandra O'Keefe, LDEO REU & SUNY Albany	2022-2023
Now: Ph.D. Candidate at University of Arizona	
Karine Holmes, LDEO REU & Iowa State University	2021-2022
Now: Ph.D. Candidate at University of Rochester	
Chandler Morris, Columbia University	2019-2022
Now: Ph.D. Candidate at Brown University	
<i>Undergraduate Research Assistants</i>	
Radha Carollo	2023-2024
Lianna Pront	2022-2024
Lexy Pryor	2022-2024
Vivian Jackson	2023
Bryn Stecher	2020-2021
<i>Columbia Climate School Graduate Research Assistants</i>	
Yaiza Kinney	2023-2024
Arjana Ejupi	2022-2023

PRESENTATIONS

Invited Talks

Pallone, C.T. Orbital Forcing of the Eastern Equatorial Pacific in the Late Pleistocene. 2025
Climate and Environment Lunch Bunch Series, Brown University Department of
Earth, Environmental & Planetary Sciences.

Pallone, C.T., Unraveling Paleo-Productivity in the Eastern Equatorial Pacific with U, Th, and

Pa. 2023 Climate and Paleo Seminar Series, Woods Hole Oceanographic Institution.

Selected Oral Presentations

McManus, J.F., **Pallone, C.T.**, Arellano, A., et al. Changes in Deep Circulation and Carbon Storage in the Eastern North Atlantic Ocean Across the Mid-Pleistocene Transition. (Presenting Author). 2026 EGU General Assembly.

Pallone, C.T., McManus, J.F., Arellano, A., Matthews, M., Alonso García, M., Lasluisa Molina, E.R. A Depth Transect of North Atlantic Deep-Water Masses at the Iberian Margin in the Late Pleistocene. 2026 Ocean Sciences Meeting.

Pallone, C.T., Jacobel, A.W., McManus, J.F. Eastern equatorial Pacific paleo-productivity did not drive global carbon cycling during the late Pleistocene. 2023 AGU Fall Meeting.

Pallone, C.T., Low Latitude Insolation Forcing of the Eastern Equatorial Pacific in the Late Pleistocene Inferred from Thermocline Reconstructions. 2023 Graduate Climate Conference.

Pallone, C.T., Eastern equatorial Pacific paleo-productivity during the late Pleistocene: insights from uranium series isotope proxies. 2023 Comer Climate Conference.

Pallone, C.T., McManus, J.F., Evaluating Orbital Forcing on the Eastern Equatorial Pacific Thermocline and El Niño-Southern Oscillation during Marine Isotope Stage 5. 2021 Comer Climate Conference.

Convened Conference Sessions

Alonso García, M., **Pallone, C.T.**, Salgueiro, E., Amarathunga, U., Standring, P., Lasluisa Molina, E. Iberian margin and Mediterranean paleoclimate and paleoceanography from a foraminifer perspective. FORAMS 2026.

Pallone, C.T., Jasper, C.J., Cargill, S.K., Gross, G. North Atlantic Paleoclimate: Sedimentary Archives of Climate, Ocean Circulation, and Ice Sheet Interactions. (Primary Convener). 2025 AGU Fall Meeting.

Fastovich, D., Anderson, C.H., Belanger, B., Kinsley, C.W., **Pallone, C.T.** Climate Reconstruction from the Pacific Region: Insights into Past Oceanic and Atmospheric Conditions. (Early Career Convener). 2024 AGU Fall Meeting.

Selected Poster Presentations

Pallone, C.T., McGee, D., Kirtland-Turner, S. Paleoredox Reconstructions from the Deep Equatorial Atlantic Ocean across Eocene Thermal Maximum 3. 2025 AGU Fall Meeting.

Pallone, C.T., McManus, J.F., Arellano, A., Matthews, M., Alonso García, M., Lasluisa Molina, E.R. Late Pleistocene benthic oxygen isotope stratigraphy and deep water mass history at IODP Site U1587 on the Iberian Margin. 2025 Gordon Geochronology Research Conference.

Pallone, C.T., Holmes, K., Morris, C.A., McManus, J.F. Orbital forcing of the eastern equatorial Pacific thermocline in the late Pleistocene. (Poster). 2024 AGU Fall Meeting.

Pallone, C.T., McManus, J.F., Jacobel, A.W. Synthesizing Late Pleistocene Paleo-Productivity Records from ODP Site 1240 with New Insights from U-Th-Pa Measurements. 2022 International Conference on Paleoceanography 14.

Co-Authored Student Presentations

Matthews, M., McManus, J.F., **Pallone, C.T.** Orbital- and Millennial-scale Deep-ocean

Circulation and Climate Variability During the Mid-Brunhes from Isotopic Proxy Records at IODP Site U1587 on the Iberian Margin. (Poster). 2024 AGU Fall Meeting.

Cumbe, C.,* **Pallone, C.T.**, McManus, J.F. Did Icebergs Drive Major Climate Changes During the Last Ice Age? (Poster). 2024 AGU Fall Meeting. (*Student Travel Grant Recipient).

Perrone, A., McManus, J.F., **Pallone, C.T.**, How did the cryosphere affect the North Atlantic Ocean during the Last Glacial Maximum? (Poster). 2023 AGU Fall Meeting.

Latting, H., McManus, J.F., **Pallone, C.T.**, Zhou, Y. Deep Ocean Circulation in the West and East North Atlantic During the Last Glacial Period and Holocene. (Poster). 2023 AGU Fall Meeting.

Kim, E., McManus, J.F., **Pallone, C.T.** North Atlantic Deep Water Masses Throughout the Mid-Pleistocene Transition: Changes at Site U1587 on the Iberian Margin. (Poster). 2023 AGU Fall Meeting.

Smith, S., McManus, J.F., **Pallone, C.T.** Investigating Past Thermocline Conditions in the Eastern Equatorial Pacific Since the Last Glacial Interval Using Oxygen Isotope Gradients from Shallow- and Deep-dwelling Planktonic Foraminifera. (Poster). 2023 AGU Fall Meeting.

McMahon, C., McManus, J.F., **Pallone, C.T.**, How Did the Mediterranean Outflow Vary In the Last 30,000 Years? (Poster). 2023 AGU Fall Meeting.

O’Keefe, A., **Pallone, C.T.**, McManus, J.F. Reconstructing the Thermocline in the Eastern Equatorial Pacific since the Last Glacial Interval. (Poster). 2023 AGU Fall Meeting.

Holmes, K.,* **Pallone, C.T.**, McManus, J.F. Multispecies foraminifera reconstruction of the Eastern Equatorial Pacific thermocline during the Holocene and last glacial interval. (Poster). 2021 AGU Fall Meeting. (*Student Travel Grant Recipient).

Morris, C., **Pallone, C.T.**, McManus, J.F. A foraminifera-derived reconstruction of the Eastern Equatorial Pacific thermocline between 70,000 and 150,000 years ago. (Poster). 2021 AGU Fall Meeting.

ACADEMIC SERVICE

Early Career Convener, 1 st New England Paleoclimate Workshop	2026
Reviewer	2025-2026
<i>Nature Geoscience, Nature Communications, Journal of Geophysical Research: Oceans, Paleoceanography and Paleoclimatology, Climate of the Past</i>	
Co-Organizer, LDEO Protactinium, Ionium, Thorium, Helium Seminar	2023-2024
Alumni & Postdoc Liaison, Graduate Student Council	2023-2024
Dept. of Earth & Environmental Sciences, Columbia University	
Student Representative, LDEO Executive Committee	2022-2023
Co-Organizer, LDEO Colloquium	2021-2023
Co-Organizer, LDEO Geochemistry Seminar	2020-2021
Co-Organizer, LDEO Geochemistry Coffee Hour	2020-2021

OUTREACH

Volunteer, Intrepid Museum Kids Week	2024
Volunteer, Barnard College Environmental Science Pathways Scholars Program	2023-2024
Ship-to-Shore Broadcaster, <i>R/V JOIDES Resolution</i>	2022
Undergraduate Mentor, Women in Science at Columbia University	2022
Volunteer, Girls' Science Day, Columbia University	2021
Outreach Volunteer, LDEO Core Repository	2019

PROFESSIONAL AFFILIATIONS

International Ocean Drilling Programme (IODP ³) Expedition 506S: Stratigraphic Integration of North Atlantic Legacy Sites	2026-Present
American Geophysical Union	2018-Present